Received: 31 March 2021 | Revised: 6 January 2022 | Accepted: 6 January 2022
DOI: 10.1002/stc.2928

RESEARCH ARTICLE

WILEY

Monitoring, intelligent perception, and early warning of vortex-induced vibration of suspension bridge

Danhui Dan[1,2] | **Houjin Li**[1]

[1]School of Civil Engineering, Tongji University, Shanghai, China

[2]Key Laboratory of Performance Evolution and Control for Engineering Structures of Ministry of Education, Tongji University, Shanghai, China

Correspondence
Danhui Dan, Tongji University, Room 709, Bridge Building, 1239 Siping Road, Shanghai 200092, China.
Email: dandanhui@tongji.edu.cn

Funding information
The Fundamental Research Funds for the Central Universities, Grant/Award Number: 20210205; Zhejiang Zhoushan Sea-crossing Bridge Co., Ltd. technology development project "Vortex Vibration Prediction Technology of Xihoumen Bridge"; National Key R&D Program of China, Grant/Award Number: 2017YFF0205605; National Natural Science Foundation of China, Grant/Award Number: 51878490

Summary

Vortex-induced vibrations (VIVs) are a serious problem in suspension bridges and other long-span bridges during their service periods. It can cause excessive amplitudes of the structure under low wind speeds, not only affecting driving comfort and safety but also increasing the risk of fatigue failure. Previous research on the identification and evaluation of bridge VIV events during their service periods was based on offline batch processing and analyses of monitoring data, which cannot realize real-time perception, calculation, and early warning online. This paper proposes an intelligent monitoring and warning method for VIVs of suspension bridges based on the recursive Hilbert transform according to the vibration characteristics of single-mode sinusoidal-like vibrations of engineering structures during VIVs. First, the real-time acceleration integral algorithm is used to realize the real-time calculation from the acceleration monitoring data to the dynamic displacement of the stiffening beam. Then, the recursive Hilbert transform is used to obtain real-time analytical signals of the structural displacement during VIVs. Based on its single-mode near-circular trajectory characteristic, the VIV index and the real-time analysis method are proposed to characterize the development trend of VIV events. This online extraction algorithm can realize the first warning and whole-process tracking and perception of VIV events. Furthermore, this paper presents a real-time online identification method for key motion parameters such as the instantaneous frequency, phase, and amplitude of the structure during VIV, laying the foundation for the real-time monitoring of the entire VIV process and further evaluation and management decision making. The accuracy, reliability, and engineering feasibility of the proposed method are verified by performing numerical simulations and monitoring VIV data of a real bridge.

KEYWORDS

acceleration integral, intelligent perception, real-time online calculation, recursive Hilbert transform, suspension bridge, vortex-induced vibration (VIV)

Danhui Dan and Houjin Li contributed equally to this work.

1 | INTRODUCTION

Vortex-induced vibrations (VIVs) result primarily from periodic vortex separation when air flows through the bluff body section. In the initial stage of VIVs, the vortex separates and reacts to the structure, and when the vibration achieves a certain threshold, the nonlinear self-excited force caused by the fluid–structure interaction significantly impacts resonance. For suspension bridges and other cable-supported bridges, an increased span will lead to an increased structural flexibility and a decreased damping ratio, making them more prone to VIV events.[1,2] Generally, this type of single-mode large-scale vibration occurs in a low wind speed range of 6–12 m/s. Although VIVs normally do not lead to structural collapse, they may cause serious fatigue problems with respect to key components of the engineering structure.[3] For bridge structures, VIV directly affects driving comfort and safety. VIV events have occurred in many cable-supported bridges in recent years, such as the Yi Sun-sin Bridge in South Korea, the Xihoumen Bridge in China, and the Great Belt Bridge in Denmark,[2,4,5] and this kind of event often causes widespread social concern. However, based on the cognition that bridge VIVs are classified as limited-amplitude vibrations, this type of vibration is allowed to occur in the bridge design process, and it is mainly controlled by limiting the maximum amplitude. When vortex vibration occurs in a bridge, the common response is to block traffic and then control the vibration. This lagging treatment method cannot meet the concerns of the public or the management needs of owners. Therefore, if real-time online monitoring, perception, and early warning of bridge VIV events can be realized, it will not only address the practical concerns of users, owners, and managers but also provide direct support for traffic management decision making in the traffic management department.

Recent research on bridge VIV mechanisms is mainly carried out using wind tunnel experiments and numerical simulations based on computational fluid dynamics (CFD). The purpose of the bridge VIV wind tunnel experiment research is to explore the bridge VIV self-excited force model, evaluate the amplitude of the bridge VIV, and determine the influence of the main girder section geometry on VIV[6,7]; however, owing to the size effect, it may be difficult to accurately simulate the turbulence field and vortex shedding excitation of real bridge VIV events in wind tunnel experiments, making it difficult to accurately predict the vibration response of real bridges.[8] The CFD method makes it difficult to accurately determine the initial flow field condition and structural parameters, and it is also difficult to simulate the influence of time-varying damping, which causes the results of CFD modeling and simulation based on the Reynolds number to deviate from reality.[9] In summary, wind tunnel tests and CFD are still semitheoretical and semiempirical methods that are not suitable for the identification and early warning of a real bridge VIVs.

Presently, many long-span bridges have established structural health monitoring systems, which enable the real-time acquisition of structural response, environment, load, and other data during the bridge operation. However, there has been little research into the use of monitoring information for real-time online identification of the abnormal operation status of structures. In recent years, some suspension bridge VIV events have been recorded by bridge-mounted monitoring systems.. Therefore, there has been research on the use of monitoring data to identify the VIV characteristics and regularity. Li et al.[10,11] used cluster analysis of wind environment monitoring data collected by the Xihoumen Bridge Health Monitoring System to identify wind speed and direction data during VIVs and analyzed the relationship between the wind velocity field and bridge VIV; Xu et al.[7] also analyzed the data of acceleration and wind field obtained from the monitoring system of Xihoumen Bridge, established the distribution model of the average wind speed and wind direction within the bridge site during VIVs, and proposed the RMS index and ratio index of resonance frequency based on the acceleration monitoring data to distinguish VIV events. Based on their research, Cao et al.[12] tried to predict VIVs with a fixed evaluation standard of RMS of acceleration set to 5 cm/s^2; Huang et al.[6] used the random decrement method to process real bridge monitoring signals, defined the peak coefficient of variation (COV) of the processed signal as the characteristic index, calculated the index based on the long-term monitoring data, and established the statistical quantile value as the threshold value to establish the VIV criterion.

These works are all based on monitoring data acquired during the VIV period, which are verified and screened after the event, and they are all statistically analyzed by batch processing after the event. The quality of the relationship model based on this is subject to the quality of the monitoring data with the VIV label. At the same time, the identification methods of VIV events given by these studies have not been tested by blind measurement in advance, and some defects are as follows: (1) The distinguish index is calculated based on the statistical analysis of batch data, which cannot realize the real-time state perception of the whole process of VIV events and the timely alarm of VIV; (2) the basic kinematic and dynamic parameters of the bridge (bridge instantaneous frequency, instantaneous phase, vibration amplitude, etc.) cannot be monitored in real time; (3) it is unable to monitor and reproduce the vibration attitude of the whole bridge in real time; and (4) the VIV judgment criteria, proposed by previous studies for specific bridge VIV events, specific location acceleration data, and specific modal frequencies, lack universal applicability to all types of bridges. Owing to the above shortcomings, this kind of research work is still at the theoretical level and is not suitable for real bridge VIV monitoring system and cannot be used to solve the problems of online monitoring, perception, or the early warning of VIV events.

Therefore, this paper presents an online monitoring, intelligent perception, and real-time warning scheme for VIV events, which can be widely used in various engineering structures. First, the real-time recursive acceleration integral algorithm proposed by Zheng et al.[13] is used to realize the online displacement monitoring of a real bridge, and the analytical signal of VIV displacement and its complex plane trajectory are obtained using the established real-time recursive Hilbert transform method Based on this, the VIV event characterization index is proposed to realize the identification, early warning, and tracking of VIVs, as well as the real-time high-precision measurement of the whole-process VIV parameters. The effectiveness of the proposed scheme was verified using simulation examples and real bridge monitoring data.

2 | MONITORING AND PERCEPTION BASED ON CHARACTERISTICS OF VIV SIGNALS

2.1 | Characteristics of structures VIV response signals

VIV is a wind-induced vibration phenomenon occurring in structures. For long-span bridges, periodic vortex shedding occurs when a steady airflow passes through the bridge section. VIVs may be excited when the frequency of vortex shedding is close to the bridge's natural frequency. The VIV of a bridge has the following characteristics: (1) VIV is usually a limited-amplitude vibration at low wind speed; (2) when the excitation frequency is close to the natural frequency of the bridge, VIV locking will occur, making the bridge vibrate greatly; (3) the vibration amplitude is related to the bridge section shape, damping, mass, and Strouhal number; and (4) vortex shedding can also cause bending and torsion vibrations.

In order to explore the vibration laws of bridges during VIV, previous researchers have proposed many mechanical models of bridge VIV based on numerical simulations, wind tunnel tests, and engineering experience. Models describe the across-flow response of a rigid sectional model of a bridge deck mounted on linear springs and submerged in a fluid stream of uniform velocity U. This system has only one vertical vibration degree of freedom; therefore, the vertical vibration of the bridge during VIV can be described by the following dynamic equations:

$$m\ddot{u}+c\dot{u}+ku=F_y, \tag{1}$$

where u is the vibration displacement, m is the structural mass per unit length, c is the structural damping, k is the linear stiffness of the structure, and F_y is the vertical self-excited force of the bridge. According to previous studies,[14–16] it can be expressed as follows:

$$F_y=\frac{\rho D^2}{m}\left[Y_1(k)\cdot\left(1-\varepsilon(K)\ddot{u}^2-\varepsilon_t w_r^2\right)\cdot\dot{u}+Y_2(k)u\right], \tag{2}$$

where ρ is the fluid density; D is the height of the bridge section; U is the wind velocity; and $Y_1(k)$, $Y_2(k)$, and $\varepsilon(K)$ are statistical parameters based on the wind tunnel tests. ε_t is the nonlinear damping of the bridge under VIVs, and $w_r=w(t)/U$, which is a dimensionless vertical turbulence term.

Based on various kinds of vortex-induced force models and engineering experience, during VIVs, depending on its frequency-locking characteristics, the vibration displacement u of the bridge can be approximated as a narrow-band harmonic model[1]:

$$u(t)=a(t)\sin(\omega t+\phi)+noise, \tag{3}$$

where t is the time, ω is the frequency of bridge vibration during VIV, ϕ is the phase, and $a(t)$ is the VIV amplitude.

According to Khalak and Williamson,[17] the amplitude coefficient $a(t)$ of VIV is affected by the phase difference between the structural displacement response and the vortex-induced force, which is an amplitude-modulated signal that changes slowly with time and can be written as

$$a(t) = a_0 + \sum_{m=1}^{M} a_m \, cos(\omega_m t + \phi_m), \tag{4}$$

where a_0 is the initial amplitude; a_m, ω_m, and ϕ_m are the amplitude, frequency, and phase of the mth amplitude modulation signal component, respectively; and M is the number of amplitude modulation signal components. Under the condition of single-mode narrow-band signal, $\omega_m \ll \omega$.

When VIVs occur, the structural vibration response approximately presents the characteristics of a single-frequency variable-amplitude harmonic signal; most of the vibration energy is concentrated at a certain frequency, and the spectrum value is much larger than the other frequencies, which is very similar to the spectrum of the single-mode vibration signal. In contrast, there are many modal components in the non-VIV displacement response of the structure under random environment excitation. The vibration signal is nonstationary, time varying, and random, and it behaves as a multimodal random vibration. According to this difference, the single-mode signal characteristic of the structural vibration response caused by VIV can consequently be used as the distinguishing condition of VIVs and non-VIVs to monitor and identify VIV events.

2.2 | Definition of monitoring and perception

As mentioned above, the vibration characteristics of the bridge under a normal state are significantly different from those under VIV. Therefore, the monitoring and perception of bridge's VIV events can be distinguished according to this characteristic. The first is the monitoring process, where the acceleration response is measured in real time using a set of acceleration sensors preinstalled on the stiffened girders of the suspension bridge, and then, the real-time data are transmitted and stored in the central monitoring server of the bridge health monitoring system.

Perception is the online processing of the acquired monitoring data to identify the deeper meaning behind the data and includes qualitative perception and quantitative perception. The first is the perception of the "nature" of the whole process of the VIV event, including the instant detection of the emergence and decay of the VIV, with the aim of realizing early warning or removing the alarm for the bridge owner and the vehicles on the bridge; and the real-time tracking of the evolution process when the VIV occurs, including the growth period (gradually increasing amplitude), the stable period (constant amplitude), and the declining period (gradually decreasing amplitude). The second is the perception of the "quantity" of the parameters of VIV events, including the identification of the current state quantities, such as vibration response and modal parameters, and also the real-time monitoring of the kinematics, dynamics, and other related mechanical parameters of the bridge during VIV.

In this way, the VIV of the bridge can be evaluated and intervened, as shown in Figure 1.

2.3 | Qualitative perception of VIV events based on VIV index

To realize the whole-process perception of the "nature" of VIV events, it is necessary to define a VIV index to distinguish VIV from non-VIV events, mark the whole process of VIV events, and quantify the VIV strength. This VIV index can be defined according to the single-mode characteristics of the vibration response signal.

We perform the Hilbert transform on signal $u(t)$ as follows:

$$v(t) = H(u(t)), \tag{5}$$

where $H(\cdot)$ is the transform operator.

Then, the analytic signal $z(t)$ can be obtained as

$$z(t) = u(t) + iv(t) = A(t)e^{i\Phi(t)}, \tag{6}$$

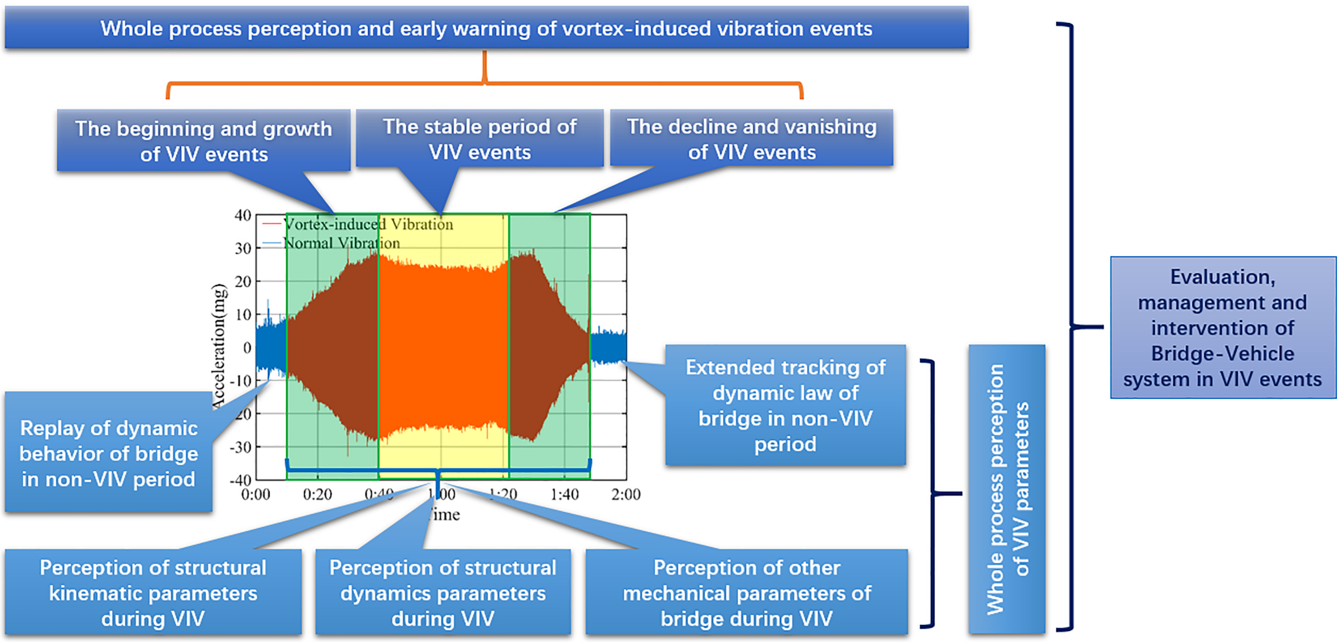

FIGURE 1 Comprehensive monitoring and perception of bridge VIV events

where i is the imaginary unit; its amplitude and phase are $A(t)$ and $\Phi(t)$, respectively, which are given by

$$A(t) = a(t) = a_0 + \sum_{m=1}^{M} a_m \cos(\omega_m t + \phi_m), \tag{7}$$

$$\Phi(t) = \omega t + \phi(t). \tag{8}$$

Under the condition of $\omega_m \ll \omega$, the radial variation of the trajectory of the analytical signal $z(t)$ is much less than the circumferential variation, so its trajectory approximately presents a circular motion in a short time; if this condition is not met, the trajectory of $z(t)$ will present a chaotic shape far from the circle. Therefore, the VIV index can be defined as the ratio *Rratio* of the minimum amplitude $A_{\min}$ to the maximum amplitude $A_{\max}$ of the analytical signal at a given sampling time T_s:

$$Rratio = \frac{A_{\min}}{A_{\max}}. \tag{9}$$

Generally, assume that there is only one amplitude modulation signal component.

$$Rratio = \frac{A_{\min}}{A_{\max}} = \frac{a_0 - a_m}{a_0 + a_m}. \tag{10}$$

It can be seen that the index *Rratio* is between 0 and 1. Within the specified time T_s, the closer *Rratio* is to 0, the farther is the signal from the single-mode signal characteristics, and the closer *Rratio* to 1, the more obvious is the single-mode feature of the signal.

Considering the general characteristics of VIV response signals, on the one hand, the signal is single mode, which means that only one mode has a high signal-to-noise ratio, and the other modes are submerged in the noise level; on the other hand, the amplitude change rate of the single mode is much less than the phase change rate. Therefore, the trajectory of Hilbert's analytic signal of the vortex response is approximately circular in one period. In the case of non-VIV, the signal contains multiple modal components and large noise interference, and the trajectory of the analytical signal will be far from the circular shape. Two simple numerical examples are shown in Figure 2. The trajectory of the

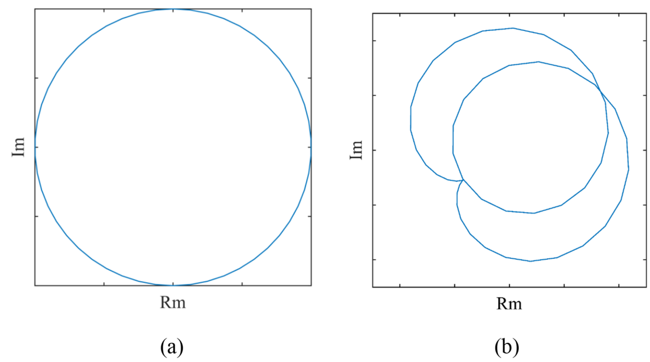

FIGURE 2 Trajectories of ideal analytic signals. (a) Single mode and (b) multimode

ideal analytical single-mode tuning signal is shown in Figure 2a, and Figure 2b shows the trajectory of the tuning analytic signal containing the two modes. It can be seen from the figure that in one period, the single-mode trajectory presents a standard circle, whereas the multimode trajectory has an irregular shape.

It can also be seen from Figure 2 that in a shorter sampling time (less than one period), the trajectory of the ideal analytical single-mode signal approximately follows a circle, and the VIV index *Rratio* given by Equation 9 has good single-mode characterization ability. The *Rratio* value of the theoretical single-frequency constant-amplitude sinusoidal vibration is 1, while the ratio value of the multimode signal is much less than 1, the trajectory changes significantly with time, and there is no regular pattern for the multimode signal.

In the actual state of bridge VIV, although the vibration response presents a single mode owing to the frequency-locking effect, there is still interference of other modes and noise, so the VIV index *Rratio* fluctuates slightly around a number less than 1; in the case of non-VIV, the bridge vibrates under the coupling influence of traffic load and environmental random vibration, and the vibration response signal is multimode under the background of broadband noise, and its *Rratio* should fluctuate at a low level close to 0. Therefore, the VIV and non-VIV states can be distinguished by setting a multilevel threshold, and the VIV strength characteristics can also be qualitatively determined according to the value of the index.

2.4 | Quantitative perception of VIV events

In addition to the qualitative understanding of VIV events, the quantitative perception and measurement of various parameters describing VIV events are also requirements for VIV control and safety warning. It is necessary to further identify these parameters based on the acceleration vibration monitoring data on the main girder of the suspension bridge.

The acceleration is a parameter that indicates the strength of vibration, but from an engineering perspective, researchers prefer to obtain the dynamic displacement value of vibration, which belongs to the category of kinematic parameters. The technology of direct monitoring dynamic displacement includes the camera measuring method, GPS observation method, and linear variable differential transformer (LVDT). Because of their shortcomings, such as insufficient measurement accuracy, high cost, poor real-time performance, and the need for fixed displacement observation reference points, these methods cannot meet the displacement monitoring requirements of long-span bridges. Another method of estimating the amplitude of the VIV displacement was carried out in conjunction with wind tunnel tests. The vortex excitation force of the bridge can be estimated by the wind tunnel test, and then, the single-degree-of-freedom dynamic equations of the bridge section can be developed to obtain the amplitude of the bridge VIV.[18,19]

For long-span bridge health monitoring systems, it is more common to choose acceleration sensors for vibration monitoring. The structural displacement of the VIV can be obtained by the acceleration integral. This method can be divided into two categories: frequency-domain integration and time-domain integration. The frequency-domain integration results in a large truncation error because of the Fourier transform and inverse transform operation. The time-domain integration method directly processes the acceleration signal, but there remain problems of baseline drift and noise interference, and the acceleration integration algorithm based on signal batch processing cannot realize the online real-time VIV measurement. Therefore, an online time-domain acceleration integration algorithm is needed to monitor VIV displacement. The key to VIV monitoring will be discussed in detail later.

In addition to the dynamic displacement of the measuring points, it is also necessary to monitor the movement posture parameters of the entire bridge in real time. For long-span suspension bridges, real-time monitoring of the torsional motion of the cross section is important with respect to structural safety. Therefore, it is necessary to simultaneously arrange three vibration acceleration sensors in each control section of the main girder of the suspension bridge, two of which are arranged on both sides and the other is a transverse acceleration sensor. Using online acceleration integration technology, the real-time synchronous displacement in the corresponding direction of each measuring point is obtained, and the translational, pitching, and torsional angles of each section are then obtained through the geometric relationship, forming the monitoring of the motion attitude of each section, and the curve interpolation method can realize the real-time monitoring of the longitudinal motion attitude of the entire bridge during the VIV.

In addition to the kinematic parameters of VIV events, the description parameters of VIV events also include the amplitude, phase, and locking frequency. In fact, the dynamic displacement signal obtained by integration can be transformed by the Hilbert transform to obtain the analytical signal of the displacement response, and the above parameters can then be calculated using the properties of the analytical signal. Specifically, $u(t)$ and $v(t)$ are the real and imaginary parts of the analytical signal corresponding to the integral displacement signal, and the instantaneous phase $\Phi(t)$ of VIV is the reciprocal of the tangent of $u(t)$ divided by $v(t)$; the instantaneous frequency $f(t)$ can be determined by calculating the first derivative of the instantaneous phase with respect to time, and the real-time amplitude A_t of the bridge during VIV can be obtained by calculating the modulus of the resolved signal in the complex field.

2.5 | Online monitoring and perception scheme for VIV events

In fact, both the integral algorithm of acceleration and the Hilbert transform algorithm are batch processing algorithms, which require a certain length of acceleration signal sampling. In other words, the batch processing algorithm not only has poor real-time performance but also has a large number of calculations, which is not suitable for online monitoring environments and cannot meet the requirement of real-time online warning of VIV events.

To realize online monitoring, perception, and early warning of VIV events of suspension bridges, it is necessary to design a set of data processing schemes for real-time acceleration to calculate the VIV index and parameters and to realize the target of early warning and whole-process tracking of VIV events based on the index. The core of this data processing scheme is to replace the batch acceleration integral algorithm and the Hilbert transform algorithm with the online recursive algorithm, which has advantages of real time and high efficiency. Figure 3 shows a diagram of this data processing scheme.

In the online data processing scheme in Figure 3, the entire process can be divided into three main functional modules:

i. Real-time acceleration integral module;
ii. Qualitative perception and early warning module;

iii. Quantitative perception module of the whole-process parameters of VIV events.

The real-time acceleration integral module is used to realize the real-time online calculation of dynamic displacement. The calculation frequency of the displacement signal can be the same as the acceleration sampling frequency, or it can be an integer numerator. This module can be realized using the real-time acceleration integral algorithm proposed by Zheng et al.[13] In this method, the recursive least-square method is used for baseline correction, a recursive high-pass filter is used to filter the low-frequency noise in the monitoring acceleration signal, and the acceleration is then integrated. Zheng et al. verified the effectiveness of the algorithm in many ways. Therefore, this paper focuses only on the parameter configuration and performance of the algorithm when performing VIV monitoring.

Module 2 is responsible for the recursive Hilbert transform of the real-time displacement signal. The complex plane analytical signal of the dynamic displacement is obtained, and the VIV index *Rratio* is calculated within the specified time; finally, the VIV is assessed online based on the index. The short-time recursive Hilbert algorithm is the core of this module, which will be introduced in the next section.

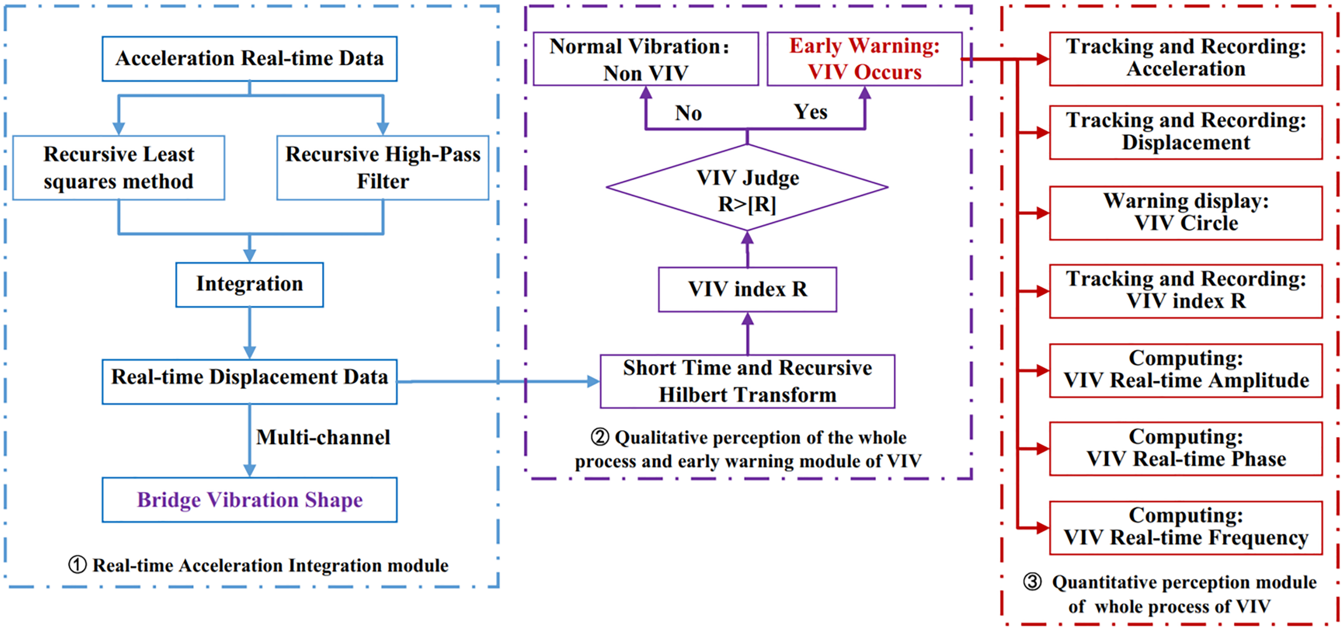

FIGURE 3 Diagram of the real-time online data processing scheme for bridge VIV events

Module 3 calculates all types of parameters of VIV events online and in real time, including kinematic and dynamic parameters. In this module, the real-time acceleration and displacement are displayed, and the trajectory of the analytic signal in the complex plane and VIV index *Rratio* are tracked in real time; the real-time frequency, phase, and amplitude of the bridge during VIV are further output. As the key indicators of the entire algorithm system, these parameters are drawn into images in real time, making it easier for bridge managers to control bridge VIV. The specific online calculation method is introduced in the next section.

3 | KEY TECHNOLOGIES OF VIV REAL-TIME MONITORING

3.1 | Acceleration integral algorithm and key parameters

To understand the real-time displacement and the motion attitude of the bridge during VIV, it is necessary to integrate the real-time acceleration of the synchronous multipoint acceleration monitoring signals of the bridge. As mentioned above, the real-time acceleration integration algorithm proposed by Zheng et al.[13] was adopted. First, the sampling frame and calculation frame of acceleration integration must be determined. The length of the sampling frame depends on the real-time requirement of the displacement result, which can be taken as an integer multiple of the sampling period of the acceleration sensor. The length of the calculation frame depends on the calculation accuracy and computing ability of the computer equipment. The longer the calculation frame, the higher is the quality of the calculation result. In the integration process, the total length of the calculation frame is unchanged and updated dynamically by adding the latest acceleration sampling frame to the calculation frame and discharging the oldest sampling frame. The algorithm first uses the least-square method to fit the potential baseline of the calculation frame and correct it. Then, it uses the high-pass filter to eliminate the low-frequency noise and finally performs time-domain integration. By repeating the above operations twice, the velocity and displacement for a sampling frame having the same length can be obtained in turn, and real-time online integration based on the acceleration monitoring data can be realized.

The key to the implementation of this method is to eliminate the noise in the acceleration monitoring signal and reduce the manual intervention as much as possible to achieve real-time online performance of the algorithm. According to Kanamori et al.[20] and Shanks,[21] the following recursive high-pass filter is selected:

$$y_j = \frac{1+q}{2}\left(x_j - x_{j-1}\right) + q y_{j-1}, \tag{11}$$

where x_j and $y_j (j = 1,2,3...)$ are the input and output signals, respectively, q is the filter parameter, and $q \in (0,1)$.

The transfer function amplitude $H(\omega)$ of the filter can be expressed as

$$H(\omega) = \frac{1+q}{2} \cdot \frac{1-e^{-i\omega\Delta t}}{1-qe^{-i\omega\Delta t}}, \tag{12}$$

where ω is the filtering frequency and Δt is the sampling time interval. For the sensor equipment of the bridge health monitoring system, this value is a fixed value, and e is the natural constant.

To ensure the accuracy of integration, the value of $H(\omega)$ should be as close as possible to 1; usually, $|H(\omega)|$ is chosen within the range of 0.97–0.99; therefore, the effect of the filter is mainly controlled by the parameters ω and q. The filter parameter q can be determined by setting the filter cutoff frequency, ω_c. Then, we use Equation 6 to eliminate low-frequency noise online. For a long-span suspension bridge, the first-order natural frequency f_s of the bridge can be calculated to determine the filter cutoff frequency f_c:

$$f_c = \alpha f_s, \tag{13}$$

where α is the filtering proportion coefficient, which can be set in the range of 1/5–1/3 for long-span bridges. Within this range, the filter can completely remove the low-frequency noise in the vibration signal while retaining the structural vibration information.

3.2 | Real-time recursive Hilbert transform and overcoming its end effect

Taking the displacement data $u(t)$ obtained by integration as the real part and its Hilbert transform $v(t)$ as the imaginary part, the analytic signal is given as

$$z(t) = u(t) + iv(t), \tag{14}$$

where $v(t)$ is the Hilbert transform of the time-domain signal $u(t)$, $v(t) = H(u(t))$, and i is the imaginary unit.

For discrete monitoring data, the continuous signals $u(t)$ and $v(t)$ can be expressed by the discrete forms $u(n)$ and $v(n)$, $n = 0,1,...,N-1$, where N is the length of the calculation frame.

The discrete Hilbert transform impulse response operator $h(m)$ can be obtained by the inverse transform of the discrete Fourier transform operator $H(n)$:

$$h(m) = \frac{1}{N}\sum_{n=0}^{N-1} H(n)e^{i\omega} = \frac{1}{N}\sum_{n=0}^{N-1} -i\,\mathrm{sgn}\left(\frac{N}{2}-n\right)\mathrm{sgn}(n)e^{i\omega} = \frac{2}{N}\sum_{n=1}^{\frac{N}{2}-1} \sin(\omega), \tag{15}$$

where $m = 0,1,...,N-1$; $\omega = 2\pi mn/N$, sgn(.) is the sign function.

Furthermore, the closed form of impulse response $h(m)$ can be expressed as

$$h(m) = \frac{2}{N}\sin^2(\pi m/2)\cot(\pi m/N). \tag{16}$$

Therefore, the Hilbert transform of the discrete displacement signal can be defined as

$$v(m) = u(m) \bigotimes h(m) = u(m) \bigotimes \left[\frac{2}{N}\sin^2(\pi m/2)\cot(\pi m/N)\right], \tag{17}$$

where $\bigotimes$ is the convolution operator. The convolution $y(m)$ of any discrete signal $x_1(m)$ and $x_2(m)$ is defined as follows:

$$y(m) = x_1(m) \bigotimes x_2(m) \triangleq \sum_{n=0}^{N-1} x_1(n) x_2(m-n). \tag{18}$$

Therefore, the Hilbert transform formula corresponding to each discrete signal point of the current calculation frame $\{u(n)\}, n = 0, 1, ..., N-1$ is given by

$$v(m) \triangleq \sum_{n=0}^{N-1} u(n) h(m-n). \tag{19}$$

According to the definition, the results of the Hilbert transform at time points $N-1$ and N can be obtained using

$$v(N-1) \triangleq \sum_{m=0}^{N-1} u(m) h(N-1-m), \tag{20a}$$

$$v(N) \triangleq \sum_{n=0}^{N-1} u(n) h(N-1-n), \tag{20b}$$

where $m = n+1$, which represents the input of the new sampling frame and the output of the last sampling frame. The recursive Hilbert transform can be written as

$$v(N) - v(N-1) = \sum_{n=0}^{N-1} h(N-1-n) u(n) - \sum_{m=0}^{N-1} h(N-1-m) u(m). \tag{21}$$

Therefore,

$$v(N) = v(N-1) + h(0) u_{\text{new}} - h(N-1) u_{\text{old}} + \sum_{m=1}^{N-1} [h(N-m) - h(N-m-1)] u(m), \tag{22}$$

where $h(N-1)$, $h(0)$, and $h(N-m+1) - h(N-m)$ are constants. In the calculation process, only the last calculation result $v(N-1)$ of the previous step must be saved. According to the dynamic update of the sampling frame, the recursive calculation can be completed by inputting the latest sampling frame u_{new}, discharging the value of the last sampling frame u_{old}, and dynamically updating the value of the intermediate frame $u(m)$. This is the recursive Hilbert transform algorithm based on the sampling points, as shown in Figure 4.

In the existing batch processing algorithm of the discrete signal Hilbert transform, N data points need to be generated simultaneously in one calculation, and the number of multiplication is N^3, which means that the number of multiplication operations for each data point is N^2. In contrast, the recursive algorithm based on sampling points only generates one data point at a time, which keeps the calculation frequency the same as the sampling frequency of the signal, and the number of calculations is $(N-1)^2 + 2$. When N is large, its computational requirement is significantly reduced compared with that of the batch processing algorithm. It is very suitable for an online environment that needs to ensure the quality of the Hilbert transform and real-time calculations.

The above recursive algorithm can be further improved to a block-based recursive formulation. This means that several sampling points comprise a sampling frame, and the calculation frame is composed of several sampling frames. In this method, the calculated frames are recombined by inputting the sampled frames in real time, and the signal is then calculated in each sampled frame instead of each sampled point. The advantage of this method is that it can adjust the output speed of the calculation results and the single-step calculation time to ensure that it is less than the acquisition time of the sampling frame to meet the real-time requirements.

Compared with the batch Hilbert transform algorithm and the recursive Hilbert transform algorithm, although both algorithms can calculate the analytical signal and both can be used for VIV identification and VIV index calculation,

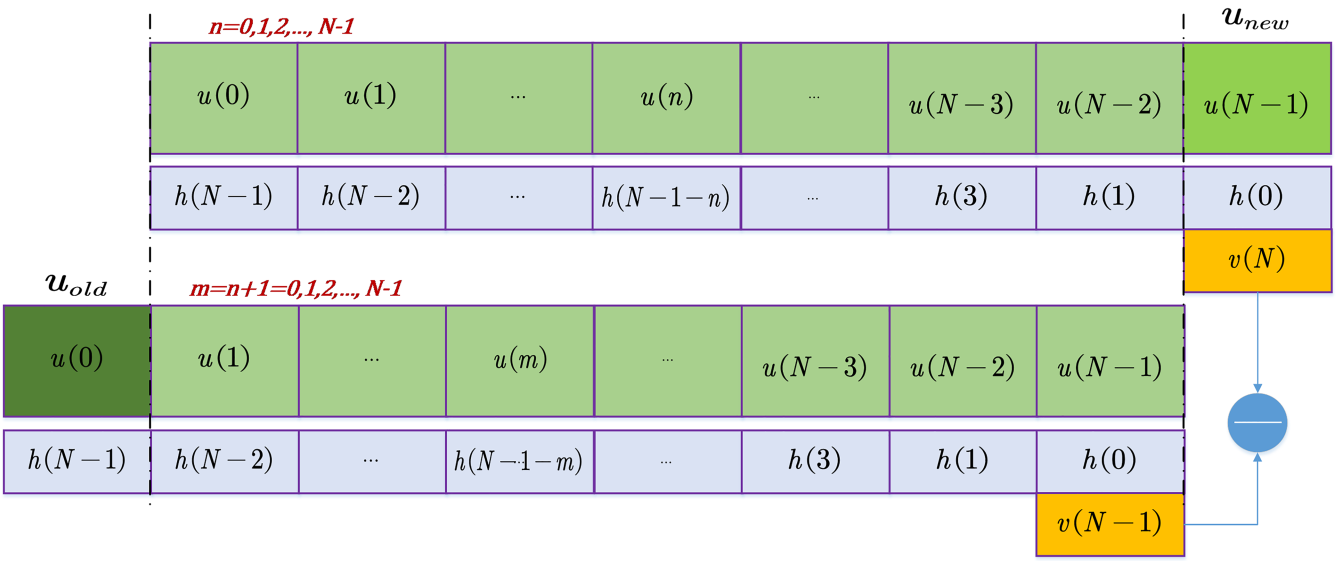

FIGURE 4 Diagram of the recursive Hilbert transform algorithm

owing to the existence of the Gibbs end effect, there remain some differences in the processing quality of the two algorithms. The Gibbs end effect means that when the Hilbert transform is carried out, the analytic signal trajectory clearly deviates from the standard circle trajectory at the end and the head parts, and its signal amplitude (the modulus of analytic signal) also fluctuates at the end.

The block-based recursive Hilbert transform algorithm can simultaneously process and weaken the end effect in each real-time calculation step. Specifically, in the initialization stage, the batch Hilbert transform is used to obtain the calculation result of the first calculation frame. Then, the new sampling frame is input, and the corresponding analytic signal is obtained by the recursive Hilbert transform. After eliminating the front-end fluctuation segment of the analytic signal, the remaining stable segment is used to connect to the iteration step to achieve dynamic and more stable updates and calculations.

4 | NUMERICAL VALIDATION

A sinusoidal acceleration signal with a frequency of 0.2268 Hz and sampling time interval $\Delta t = 0.02$ s was constructed to illustrate the feasibility of the real-time online bridge VIV intelligent perception scheme.

$$a_i = \sin(0.4536\pi \times 0.02i), i = 1,2,3\dots. \tag{23}$$

The theoretical velocity and displacement are as follows:

$$v_i = -\frac{\cos(0.4536\pi \times 0.02i)}{0.4536\pi}, i = 1,2,3\dots, \tag{24}$$

$$d_i = -\frac{\sin(0.4536\pi \times 0.02i)}{(0.4536\pi)^2}, i = 1,2,3\dots. \tag{25}$$

4.1 | Integration of ideal sine acceleration signal

For an ideal sine acceleration signal, a filtering cutoff frequency less than the first natural frequency can satisfy the filtering requirements. Therefore, the high-pass filter cutoff frequency $f_c = 0.11$ Hz is selected, recursive filter parameter

$q = 0.997$, and transfer function amplitude $|H(\omega)| = 0.975$. The integration results show that there is a time lag of 0.22 s and an amplitude attenuation of 1.8% between the integral displacement and the theoretical displacement. Because the calculation frame length of acceleration is insufficient at the beginning of the calculation, the calculation can be started by filling a large number of zeros at the front end of the data. After the initial calculation for approximately 60 s, the program can obtain stable integration results. From the calculation of the integration error (Figure 5b), it can be seen that the acceleration integration error remains at a low level after 60 s. The integral error is expressed as follows:

$$Err(t) = \frac{|D_e(t) - D_t(t)|}{\max(D_t(t))}, \tag{26}$$

where $D_e(t)$ is the integral displacement at time t after phase correction, $D_t(t)$ represents the theoretical displacement at time t, and $\max(D_t(t))$ represents the theoretical displacement amplitude of the sinusoidal acceleration signal.

4.2 | "VIV event" recognition of ideal sinusoidal signal

Noises with different signal-to-noise ratio values are added to the ideal sinusoidal acceleration signal to simulate the single-mode narrow-band acceleration response signal of the structure during VIV, which is used to test the effectiveness of the proposed VIV perception schema. First, the noisy acceleration signal needs to be integrated into the displacement signal online, and the analytical signal is then obtained using the recursive Hilbert transform, and the VIV parameters are identified according to the previous method. In the calculation, it is necessary to eliminate a large error section in the initial stage of integration and to select the integral displacement signal from 60 to 120 s for the calculation.

Figure 6a shows the trajectories of the analytical signal in the selected stage, where the coordinates of each sampling point in the figure are given by its real and imaginary parts, and Figure 6b shows an enlarged view of its partial trajectories. It can be seen from the figure that the trajectory of the analytical signal after the Hilbert transform of the standard sinusoidal signal is a standard circle, which can be used for VIV identification intuitively and accurately. Although the VIV circle extracted by the Hilbert transform has a certain offset after adding Gaussian white noise with SNR = 5, SNR = 10, and SNR = 20 to the ideal sinusoidal signal, overall, the VIV index *Rratio* is still close to 1, and the recognition error decreases with the increase in SNR. This shows that the proposed VIV index can express the characteristics of the single-mode narrow-band signal well.

4.3 | Comparison of results between recursive Hilbert transform and batch Hilbert transform

To compare the processing effect of the traditional batch Hilbert transform and the proposed recursive Hilbert transform, we selected a 120-s integral displacement signal to calculate the corresponding analytical signal when the displacement error is sufficiently small. The trajectories obtained by the two algorithms after deducting the Gibbs end

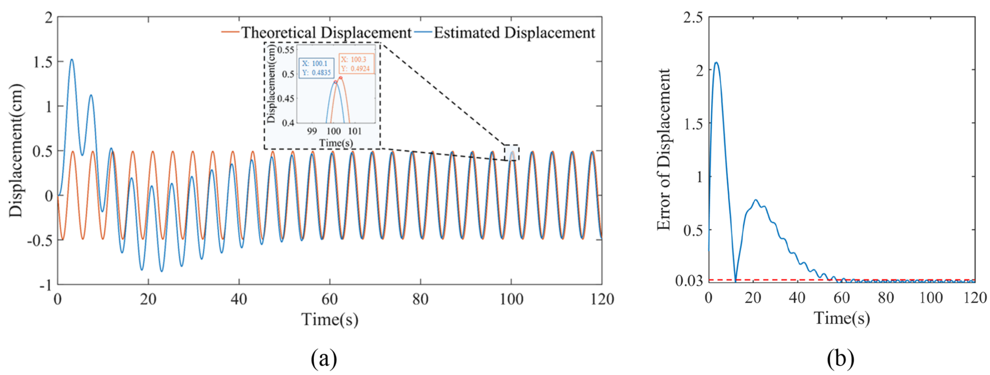

FIGURE 5 (a) Integral displacement and theoretical displacement of sinusoidal signal. (b) Displacement integral error

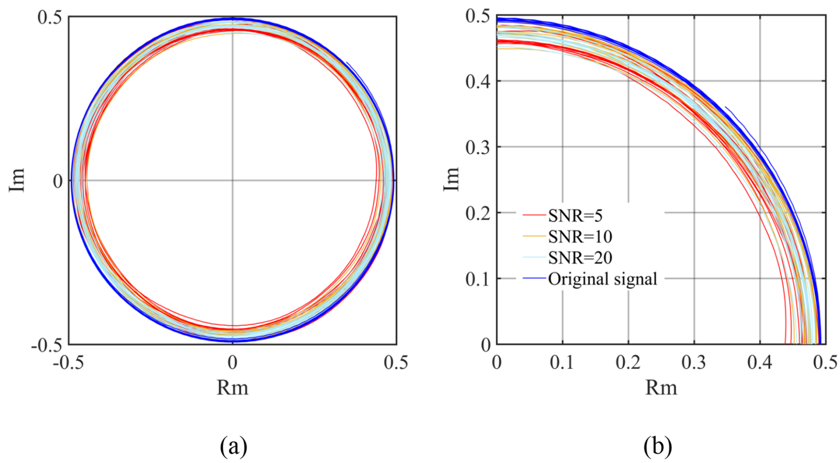

FIGURE 6 (a) Complex plane analytic signal trajectories. (b) Local trajectories

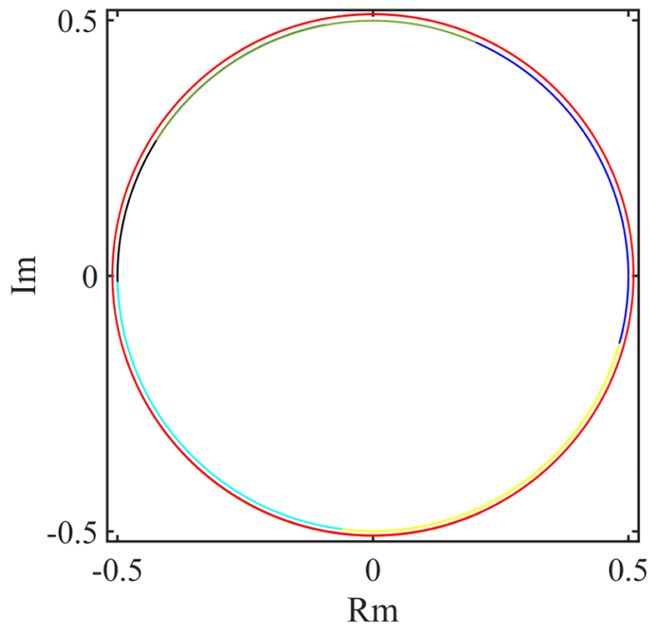

FIGURE 7 Vortex circle of batch Hilbert transform and recursive Hilbert transform

effect are shown in Figure 7. The outer red envelope circle is the batch processing method used to identify the VIV circle, and the inner multicolor VIV circle is the recursive processing result, where each color represents the input and calculation result of one sampling frame; the length of frame $N = 50$.

In Figure 7, the batch processing results are slightly enlarged to distinguish between the calculation results of the two methods. In fact, the VIV circles processed by the two methods are completely coincidental. It can also be seen from this figure that the trajectories of the analytical signals obtained by the two algorithms are all standard circular curves. This further proves that the two methods can be used to recognize the perception of VIV events. However, because of the better real-time performance of the recursive algorithm, it is more suitable for online monitoring conditions.

The time-varying amplitude of the analytical signal obtained using the two methods is shown in Figure 8. Figure 8a shows the amplitude time-history curve of the 120-s-long analytical signal obtained by the batch processing algorithm. Theoretically, the ideal amplitude of the sinusoidal signal after the Hilbert transform is 1, but the amplitude curve in Figure 8a shows a large fluctuation at the end. The fluctuation amplitude gradually decreases from the end to the middle, and the length of the significant fluctuation part at both ends accounts for approximately 75% of the total length, and a small section (approximately 30 s) curve in the middle approaches 1. The distortion of the amplitude caused by

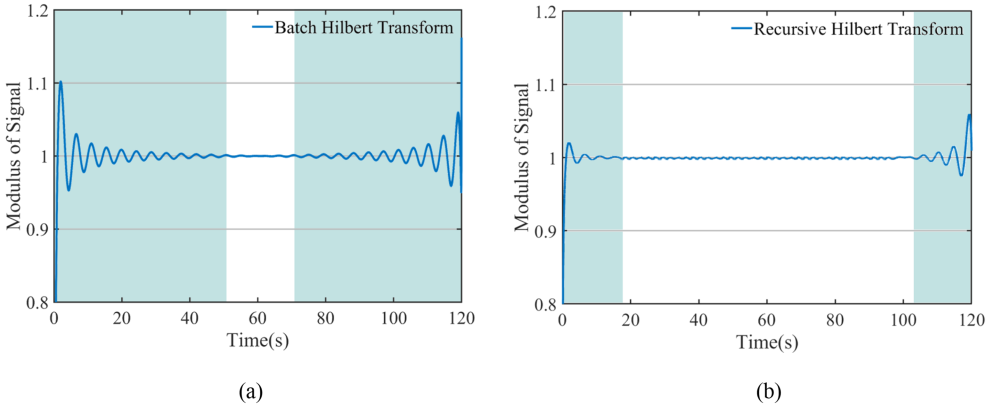

FIGURE 8 Comparison of end effects of the two algorithms. (a) Signal modulus of batch Hilbert transform. (b) Signal modulus of recursive Hilbert transform

the end effect will obviously lead to the distortion of the VIV index calculation; even worse, it will lead to a deviation in VIV identification. As can be seen, only a small section of the processing results in the middle can be used for VIV index calculations. Although the proportion of the end wave segment decreases as the signal lengthens, it will cause a rapid increase in the number of calculations, limiting its practical application.

Figure 8b shows the processing results of the module of the analytical signal using the block-based recursive Hilbert transform algorithm. By performing the recursive Hilbert transform of the ideal sinusoidal signal many times, we found that when the calculation frame length was 2000, the sampling frame length was 50, and the deletion length of the front-end fluctuation segment of the signal result after each transformation is 300. The calculation result has high accuracy, and the length of the fluctuation segment is small. As shown in Figure 8b, the module of the signal processed by the block-based recursive method fluctuates only at the beginning and end of the initial and final calculation frames. Moreover, the length of the fluctuation segment is very short, and the proportion of the fluctuation segment will continue to decrease as the data calculation length increases.

It can be seen that the block-based recursive Hilbert transform has the following advantages: (1) The high accuracy and stability of this algorithm, it can be very accurate for VIV index calculations and VIV identification regardless of the signal length; (2) the calculation error range and error size are much smaller than those of the traditional batch Hilbert transform algorithm; and (3) real-time calculation and analysis can be carried out with the continuous input of the monitoring signal. The above three advantages are very important for the real-time online monitoring of engineering structures.

4.4 | Calculation of ideal VIV signal parameters

After determining the VIVs, the parameters of the VIV were calculated. Figure 9a shows that the phase of the theoretical sinusoidal VIV signal is a regular triangular waveform, and the signal period is always constant, $T = 1/f_s = 4.4\,\text{s}$. The frequency of the vibration signal can be stably output after the initial calculation for one period, as shown in Figure 9b. This is mainly because the calculation frame for the instantaneous frequency is slightly longer than that in the acceleration integration process.

The real-time displacement signal and the standard circular trajectory in the complex plane are obtained by integrating the ideal sinusoidal acceleration signal and applying the block-based Hilbert transform, which shows that this method can identify VIV accurately and reliably. Furthermore, the real-time phase and instantaneous frequency calculation of the numerical signal also show that this method can be used to measure the VIV accurately and reliably.

5 | PRACTICAL APPLICATION

The Tiger Gate Bridge is a suspension bridge with a span of 302 + 888 + 348.5 m. A severe VIV event occurred on the bridge on May 5, 2020. After taking some measures to suppress vibration, there remained several occasional VIVs. To

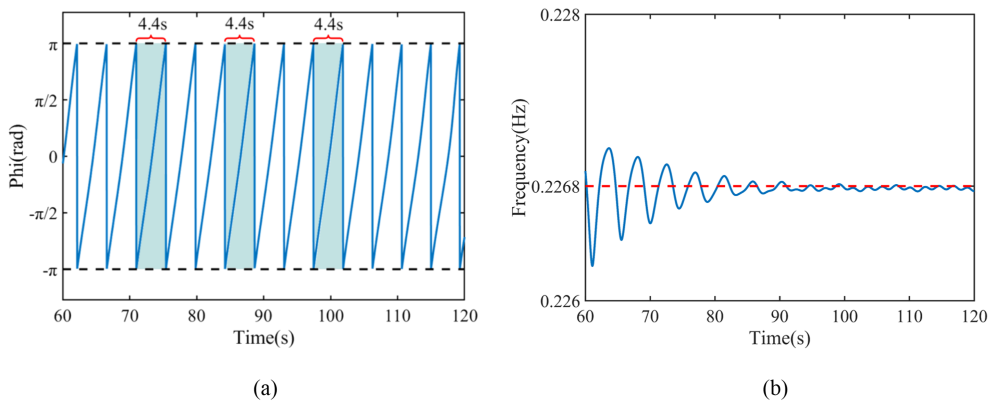

FIGURE 9 (a) Instantaneous phase. (b) Instantaneous frequency

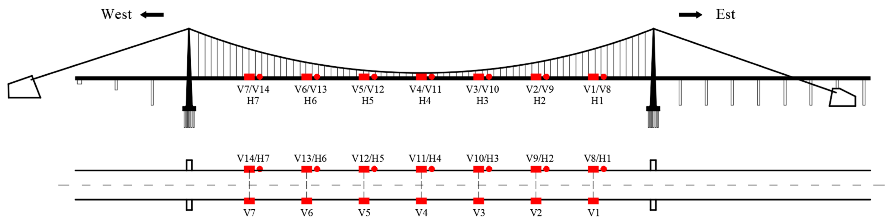

FIGURE 10 Arrangement of acceleration sensors on Tiger Gate Bridge

determine the VIV status of the bridge in real time, the owner installed a permanent health monitoring system on the bridge. The VIV parameters being monitored include acceleration, wind velocity, and wind direction. Seven bidirectional (vertical and lateral) acceleration sensors are evenly mounted on the upstream side of the main span, from V8 to V14 are vertical channels and H1 to H7 are horizontal channels; and seven vertical sensors are evenly arranged on the downstream side of the main span, numbered V1–V7, as shown in Figure 10. All sensors were sampled synchronously, and the sampling frequency was 50 Hz.

The system successfully monitored several subsequent VIV events and obtained primary structural VIV response data. These valuable monitoring data create conditions for research on long-span suspension bridge VIVs. This study attempts to use the acceleration monitoring data of the entire day on June 12, 2020, to verify our proposed monitoring, intelligent perception, and early warning schema for bridge VIV events. Based on preliminary observations, a VIV event with a large amplitude and long duration occurred from 0:00 to 2:00 on that morning.

5.1 | Measurement of real-time displacement and full bridge vibration attitude

All of the 24-h acceleration monitoring data of the day are first integrated using the real-time acceleration integration algorithm mentioned in Section 2.1. The acceleration time-history curve of the acceleration V2 channel and the Fourier spectrums of its typical stages are shown in Figure 11. It can be seen that the locking frequency of the structure during VIV is $f_s = 0.2268$ Hz, and the first-order natural frequency of the structure is $f_1 = 0.1348$ Hz. Based on these characteristic frequencies, after many attempts, the filter cutoff frequency was found to be $\omega_c = 0.1\pi$, which is smaller than the fundamental frequency of the structure; moreover, the recursive filter parameter q is set to 0.99, and the transfer function amplitude $|H(\omega)| = 0.975$. Under these parameters, the integration accuracy reached the highest level.

The real-time acceleration integration was operated on the acceleration data of the V2 channel, and the 24-h displacement time history is shown in Figure 11b. As can be seen, the multiple recursive least-squares algorithm realized the baseline correction, and the high-pass filter effectively eliminated the low-frequency noise in the original acceleration signal. The integration result has no baseline drift, which is directly reflected in the mean displacement, which is close to 0. Furthermore, from the displacement time history, we can preliminarily consider that VIV occurs from 0 and 2 h, as shown in Figure 12a, and we can roughly visualize the entire process of VIV generation, stable vibration, and

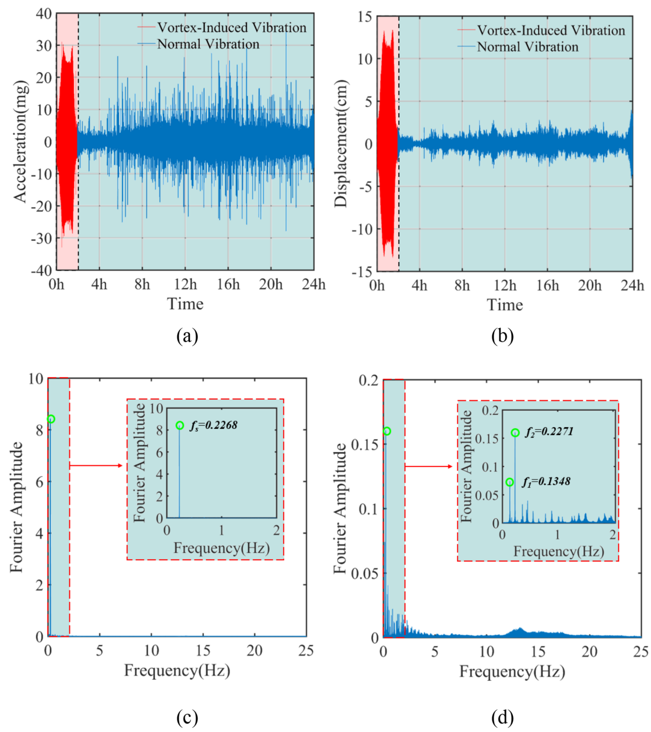

FIGURE 11 (a) Acceleration time-history curve of V2 channel. (b) Displacement time-history curve of V2 channel. (c) Fourier spectrum during VIV. (d) Fourier spectrum during non-VIV

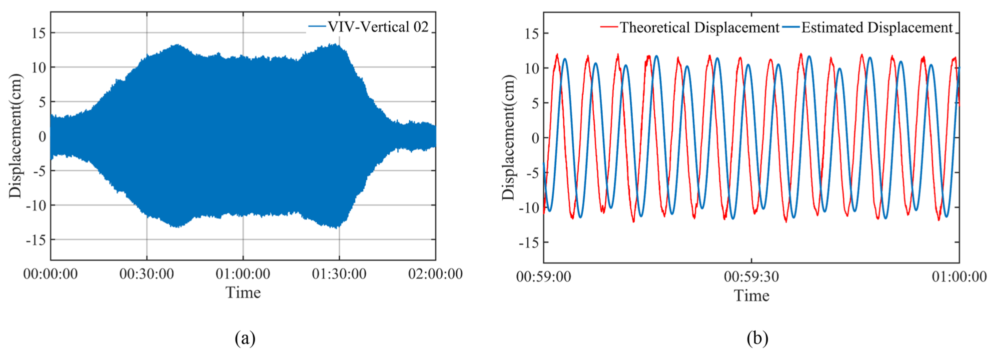

FIGURE 12 (a) Displacement time-history curve during VIV period. (b) Comparison of integral displacement and theoretical displacement during VIV stability period

attenuation. More specifically, the acceleration amplitude of the vortex stabilized section was approximately 25 mg, and the integral displacement of the stabilized section was approximately 12 cm.

To compare the accuracy of the integral displacement, the displacement time history obtained by the batch frequency-domain integration method is also shown in Figure 12b, which may be called the theoretical displacement. According to the single-mode vibration characteristics of VIV, the calculation formula of displacement d obtained by the frequency-domain integration of batch processing is as follows:

$$d = \frac{a}{4\pi^2 (f_s + \Delta f)^2}, \tag{27}$$

where a is the acceleration signal of VIV, f_s is the locking frequency, and Δf represents the other interference frequency components.

Figure 12b shows that the integer value is consistent with the theoretical value for the entire vibration process, which indicates that the integral result retains the spectrum structure of the original vibration signal well. There is a small difference in the displacement amplitude, which is mainly because the fixed frequency $f_s = 0.2268$ Hz is used to calculate the theoretical displacement, but during an actual bridge vibration, the change in aerodynamic stiffness and damping will constantly change the structure frequency. At the same time, besides this fixed frequency, the signal inevitably has some other interference frequency components, which results in the theoretical value being larger than the integral value. In addition, there is a stable phase difference between the theoretical displacement and integral displacement, but it does not affect the VIV identification and vibration parameter calculation. In summary, the online acceleration integration algorithm adopted in this study is not only suitable for online monitoring environments with good real-time performance but also has high accuracy and a small phase lag, which is significant for VIV monitoring, intelligent perception, and providing early warning for suspension bridges.

5.2 | VIV index and trajectory of analytical signal

Using the recursive Hilbert transform algorithm, the acceleration and displacement signals of the V2 channel were processed, and the analytic signals were obtained. The analytic signal segment (00:59:00–01:00:00) in the VIV stage and the analytic signal segment (02:19:00–02:20:00) in the non-VIV stage were chosen, and their respective trajectory circles were drawn, as shown in Figure 13.

Figure 13 shows that trajectories in complex planes derived from the recursive Hilbert transform during VIVs present obvious circular characteristics, while trajectories in the non-VIV period are disordered and are nearly fully filled in the area of the circle. Another interesting phenomenon is that there are obvious differences in the shapes of the

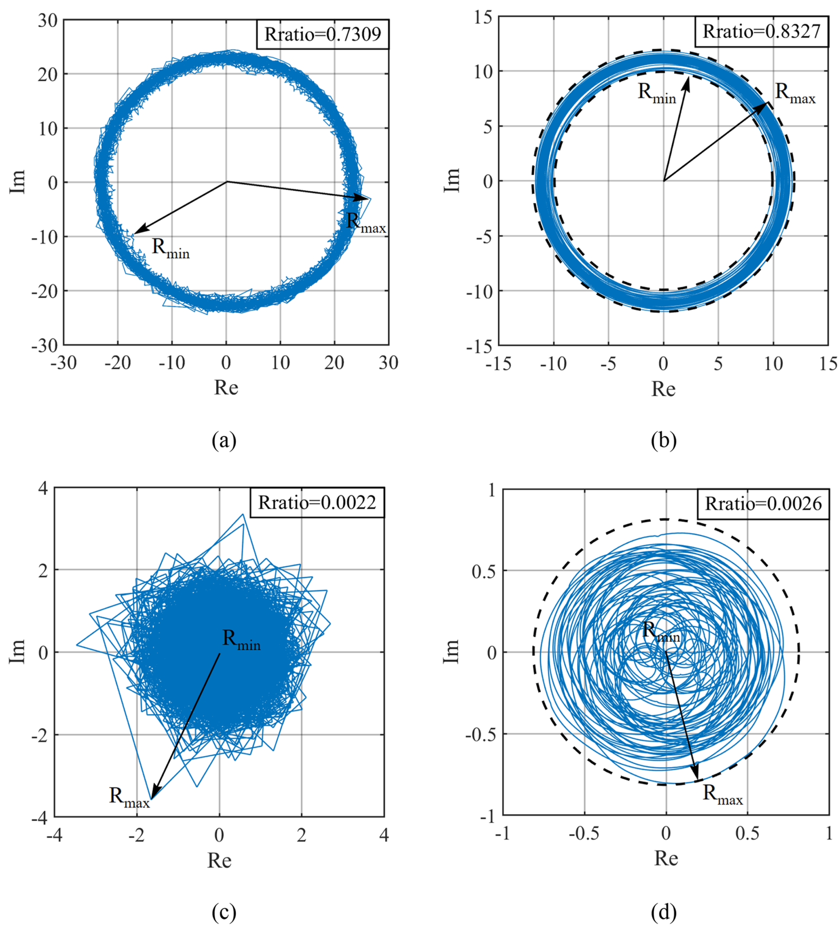

FIGURE 13 Trajectories of VIV analytical signal obtained by recursive Hilbert transform processing. (a) Acceleration in VIV period, (b) displacement in VIV period, (c) acceleration in non-VIV period, and (d) displacement in non-VIV period

trajectories corresponding to different signals. The analytical acceleration signal in the VIV period has an obvious burr, and the shape also fluctuates, but the trajectory circle of the displacement analytic signal is very smooth and has obvious characteristics. This is mainly because during the real-time acceleration integration, after many high-pass filtering and baseline correction iterations, the noise level is greatly reduced, causing the corresponding trajectory of the analytical signal to appear smooth, which is more conducive to the calculation of the VIV index. For the acceleration and displacement signals in the non-VIV period, the shape difference of the trajectory corresponding to the analytical signal is also obvious; the former trajectory is zigzag and sharp, while that of the latter is smooth.

Figure 13 also presents the value of the VIV index *Rratio* in the VIV and non-VIV periods. According to the local signal of the VIV period (00:59:00–01:00:00), the VIV index was 0.8327, while that of the non-VIV period (02:19:00–02:20:00) was 0.0026. As mentioned above, the larger the VIV index, the more obvious are the VIV characteristics, and the more certain is the occurrence of vortex vibration events. Therefore, it is confirmed that there is an obvious VIV phenomenon for the period 00:59:00–01:00:00, while the value of the VIV index is very small for the period 02:19:00–02:20:00, which is defined as the environmental random vibration.

5.3 | Perception and measurement of VIV event

The entire VIV process was accurately perceived before, during, and after VIV by calculating the real-time VIV index and parameters of the VIV signal after the real-time recursive Hilbert transform. The recursive algorithm is proven to be suitable for an online monitoring environment, which is not only fast, but the end effect is also reduced, and the stability of the algorithm is better. Further integration with real-time acceleration integral algorithm into a global system can realize online real-time monitoring, intelligent perception, and early warning of VIV events, and the time lag of the system can be controlled at the second level. To ensure the real-time performance of the VIV warning and parameter calculation, it is also necessary to ensure that the total calculation time spent on the data block to be processed (i.e., sampling frame) is less than the total acquisition time of the data block. Therefore, we set the sampling frame length to 1 s to ensure real-time performance.

The acceleration data of the V2 channel between 0 and 2 h are calculated in real-time conditions in this way, and the real-time curve of the VIV index is obtained as shown in Figure 15. VIV events can be determined by setting appropriate VIV index thresholds. The two dashed lines in Figure 14 show the two-level warning threshold of the VIV index, which are 0.3 and 0.6, respectively, to realize the two-level warning mechanism of VIV. The threshold setting is based on the observation of the track circle shape of the recursive Hilbert analytic signal. When $Rratio = 0.3$, the track circle is basically circular and can be used as the first-level warning index for VIV prevention. When $Rratio = 0.6$, the recursive Hilbert circle presents a standard circle shape, which indicates that there is an obvious VIV, so it can be used as a secondary warning index for VIVs.

The rationality of the two-level warning threshold setting based on the VIV index can also be verified using various time-history tracking charts of VIV parameters. Real-time identification and tracking of these parameters can be performed after the occurrence of a VIV event warning. As mentioned above, the analytic signal at each sampling time can be obtained by the recursive Hilbert transform, and the instantaneous phase, frequency, and amplitude can then be calculated in real time. Figure 15 shows the real-time tracking diagram of the V2 channel during VIVs.

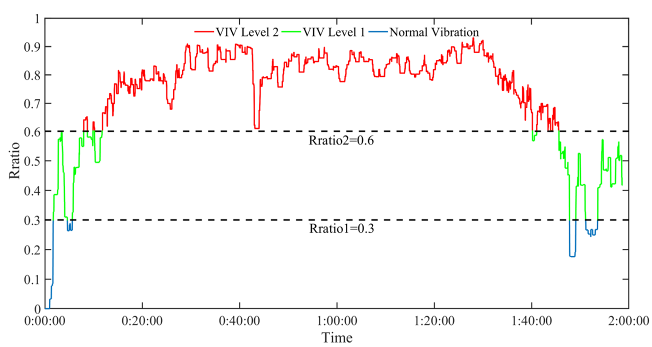

FIGURE 14 Diagram of VIV index *Rratio*

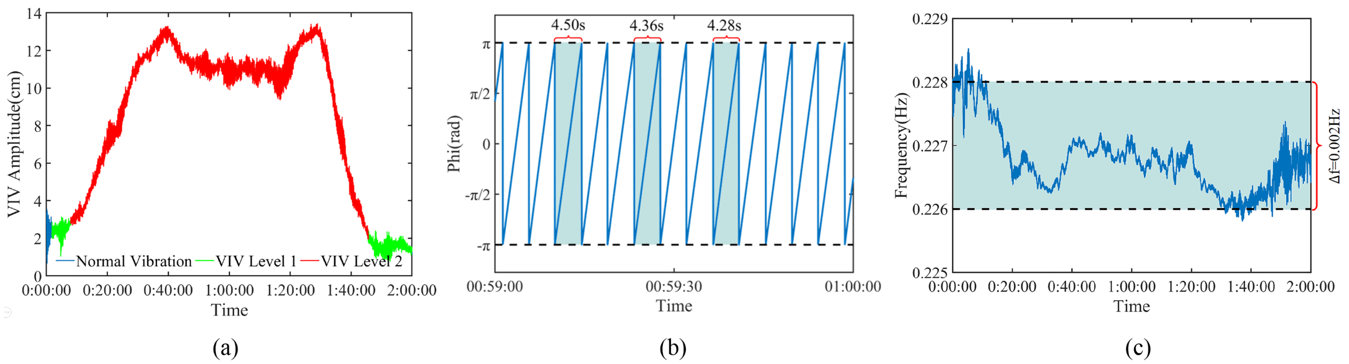

FIGURE 15 Real-time tracking and perception of VIV parameters. (a) Real-time amplitude. (b) Instantaneous phase. (c) Instantaneous frequency

In Figure 15a, the blue part of the amplitude time history is the normal vibration of the bridge, which corresponds to the situation where the VIV index is less than 0.3; the green part is VIV Level 1 of the bridge, which needs to be given the primary VIV warning in order to attract the attention of the management department; the red part is the obvious VIV of the structure, which needs to be given the secondary warning in order to control the structural vibration and take the traffic control measures in time. According to the fine perception criteria of VIV events, the accurate time period of VIV was determined from 00:07:58 to 01:45:36 on June 12.

Figure 15b shows the instantaneous phase time history of the VIV in a short time. Because the vibration signal during VIV is similar to a sinusoidal signal, its instantaneous phase also has an obvious serrated waveform, and its period is equal to the period of the locking vibration mode of VIV. In each period, the phase changes between π and π in a nearly linear manner. In fact, the locking frequency is not completely fixed during the VIV period, which is time varying. Therefore, in the phase time history, there was a slight change in the period of each vibration cycle.

Figure 15c shows the instantaneous frequency of the bridge during VIV by deriving the vibration phase. We observe that the instantaneous frequency of the bridge during VIV is not completely fixed. In this case, there is a small fluctuation of 0.002 Hz around the locking frequency during the entire VIV period. Moreover, with the development of the VIV process, this small change presents a certain law, that is, at the beginning of VIV, the instantaneous frequency is higher, and the fluctuation range is larger; as the VIV progresses, its value gradually decreases, and the fluctuation also decreases; when the VIV becomes stable, the frequency is basically stable; at the end of the VIV, the instantaneous frequency begins to produce large fluctuations. A possible explanation for this phenomenon is that the aerodynamic stiffness and damping caused by VIVs may not be constant but may have a small variation range, which requires further research.

5.4 | Discussions

Combined with the previous numerical case study and real bridge monitoring data verification, it is proven that the proposed method can be used for VIV identification and provide early warning monitoring of suspension bridges. By adopting the real-time acceleration integration algorithm, the vibration displacement monitoring of a suspension bridge girder is no longer a difficult problem, and the vibration parameters and operation state of the bridge can be comprehensively understood according to multichannel synchronous integration. Furthermore, the short-time recursive Hilbert transform can be used to identify and track the VIV indexes in real time and provide early warning of VIV events and intelligent perception of the whole process, as well as the real-time tracking and identification of various parameters during bridge VIVs, including rich vibration information such as instantaneous frequency, phase, and amplitude.

The key point of the proposed scheme is the criterion for the VIV. It should be noted that the vertical vibration amplitude of the main girder at the V2 measuring point is reflected in Figure 15a, but the VIV cannot be assessed using only this parameter. When the instantaneous amplitude of the bridge is at a low level, obvious signs of VIV can be seen from the time-history trajectory diagram of the VIV index and the VIV circle. Some of the previous studies on VIV identification can only make effective judgments when the amplitude of VIV reaches the peak value, which is too late at this time. It is meaningless to carry out bridge management and control at this time point.

According to the calculation results of the single-channel vibration monitoring signal, a real-time VIV warning with a time lag of less than seconds can be realized. More importantly, the VIV monitoring, intelligent perception, and early warning technology proposed in this paper can accurately detect the generation of VIV when the amplitude is low and when it is difficult to determine using the naked eye and body feeling and can send advanced warning information. This can provide enough lead time for bridge owners to make control decisions and implement control measures, effectively preventing the occurrence of bridge engineering accidents and the loss of public property.

Moreover, the method proposed in this paper, which is simple to apply, has low cost, high recognition frequency, and rich and intuitive results. In addition to the acceleration monitoring information, velocity, displacement, and other monitoring signals can also be used to monitor and identify VIVs. It should also be noted that based on the single-mode vibration characteristics of VIV, we can apply the method proposed in this paper to real-time monitoring and intelligent perception of VIVs of engineering structures in addition to suspension bridges, such as cables, towers, high-rise buildings, and model experiments in wind tunnel laboratories. Because of the characteristics of different structures, there may be multimode VIV, and we can continue to improve the recursive Hilbert transform method in this study by adding an online recursive band-pass filter to split the original monitoring signal into multiple single-mode sinusoidal signals, which can achieve a multimode monitoring effect. We will continue to study the identification of VIV with different structures and improve our algorithm.

6 | CONCLUSIONS

This paper proposes a real-time monitoring, intelligent perception, and early warning technology for VIV of suspension bridges based on the recursive Hilbert transform and real-time recursive acceleration integral algorithm. By performing real-time online processing of the vibration acceleration monitoring signal of the suspension bridge, VIV events can be identified and early warning provided, and some key VIV parameters can be identified and tracked synchronously and accurately. The results show that the proposed method can accurately and efficiently determine the start and end times of VIV events; accurately measure key vibration parameters such as instantaneous frequency, phase, and amplitude during the bridge VIV period; and realize the real-time online intelligent perception of the whole process of bridge VIV.

The proposed method has many functions that can not only identify and provide early warning of VIV events, but which can also sense VIV parameters comprehensively and quantitatively with high quality. Relying on the key algorithms in an online environment and the definition of the VIV index, this technology has good real-time performance, is suitable for online monitoring systems, and can provide an adequate early warning time for the control of VIV events of suspension bridges. At the same time, because the algorithm only depends on the single-mode signal characteristics of the structural vibration response during VIV, it does not need to know the specific structure, location of the measuring point, or VIV observation history in advance. This research received the attention of the Xihoumen Bridge owner and has been applied to research on this bridge. Good results were obtained without changing any algorithm parameters, and the accurate identification of VIV emergence and vibration parameters of the large-span suspension bridge verified to some extent the robustness of the proposed method. For other complex engineering structures, such as cables, towers, and high-rise buildings, VIV monitoring and intelligent perception will be verified in future.

ACKNOWLEDGMENTS

This study is funded by the Fundamental Research Funds for the Central Universities (20210205), Zhejiang Zhoushan Sea-crossing Bridge Co., Ltd. technology development project "Vortex Vibration Prediction Technology of Xihoumen Bridge", the National Natural Science Foundation of China (51878490), and the National Key R&D Program of China (2017YFF0205605).

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

AUTHOR CONTRIBUTIONS

Danhui Dan conceptualized the study, designed the methodology, conducted the investigation, supervised the study, and reviewed and edited the manuscript. Houjin Li curated the data, conducted the formal analysis, visualized the study, and wrote the original draft of the manuscript. These authors contributed equally to this work.

ORCID

Danhui Dan https://orcid.org/0000-0003-4960-1061

REFERENCES

1. Sarpkaya T. A critical review of the intrinsic nature of vortex-induced vibrations. *J Fluids Struct*. 2004;19(4):389-447.
2. Li H, Laima S, Ou J, et al. Investigation of vortex-induced vibration of a suspension bridge with two separated steel box girders based on field measurements. *Eng Struct*. 2011;33(6):1894-1907.
3. Cantero D, Øiseth O, Rønnquist A. Indirect monitoring of vortex-induced vibration of suspension bridge hangers. *Struct Health Monit*. 2018;17(4):837-849.
4. Hwang YC, Kim S, Kim HK. Cause investigation of high-mode vortex-induced vibration in a long-span suspension bridge. *Struct Infrastruct Eng*. 2020;16(1):84-93.
5. Larsen A, Esdahl S, Andersen JE, Vejrum T. Storebælt suspension bridge–vortex shedding excitation and mitigation by guide vanes. *J Wind Eng Ind Aerodyn*. 2000;88(2–3):283-296.
6. Huang Z, Li Y, Hua X, Chen Z, Wen Q. Automatic identification of bridge vortex-induced vibration using random decrement method. *Appl Sci*. 2019;9(10):2049.
7. Xu S, Ma R, Wang D, Chen A, Tian H. Prediction analysis of vortex-induced vibration of long-span suspension bridge based on monitoring data. *J Wind Eng Ind Aerodyn*. 2019;191:312-324.
8. Hua XG, Chen ZQ, Chen W, Niu HW, Huang ZW. Investigation on the effect of vibration frequency on vortex-induced vibrations by section model tests. *Wind Struct*. 2015;20(2):349-361.
9. Kang Z, Zhang C, Chang R, Ma G. A numerical investigation of the effects of Reynolds number on vortex-induced vibration of the cylinders with different mass ratios and frequency ratios. *Int J Nav Archit Ocean Eng*. 2019;11(2):835-850.
10. Li S, Laima S, Li H. Cluster analysis of winds and wind-induced vibrations on a long-span bridge based on long-term field monitoring data. *Eng Struct*. 2017;138:245-259.
11. Li H, Laima S, Zhang Q, Li N, Liu Z. Field monitoring and validation of vortex-induced vibrations of a long-span suspension bridge. *J Wind Eng Ind Aerodyn*. 2014;124:54-67.
12. Cao S, Zhang Y, Tian H, Ma R, Chang W, Chen A. Drive comfort and safety evaluation for vortex-induced vibration of a suspension bridge based on monitoring data. *J Wind Eng Ind Aerodyn*. 2020;204:104266.
13. Zheng W, Dan D, Cheng W, Xia Y. Real-time dynamic displacement monitoring with double integration of acceleration based on recursive least squares method. *Measurement*. 2019;141:460-471.
14. Xu K, Ge Y, Zhao L, Du X. Calculating vortex-induced vibration of bridge decks at different mass-damping conditions. *J Bridge Eng*. 2018;23(3):04017149.
15. Ehsan F, Scanlan RH. Vortex-induced vibrations of flexible bridges. *J Eng Mech*. 1990;116(6):1392-1411.
16. Zhu Q, Xu YL, Zhu LD, Li H. Vortex-induced vibration analysis of long-span bridges with twin-box decks under non-uniformly distributed turbulent winds. *J Wind Eng Ind Aerodyn*. 2018;172:31-41.
17. Khalak A, Williamson CH. Motions, forces and mode transitions in vortex-induced vibrations at low mass-damping. *J Fluids Struct*. 1999;13(7–8):813-851.
18. Scanlan RH (1981). State-of-the-art methods for calculating flutter, vortex-induced, and buffeting response of bridge structures (No. FHWA-RD-80-50 Final Rpt.).
19. Marra AM, Mannini C, Bartoli G. Van der Pol-type equation for modeling vortex-induced oscillations of bridge decks. *J Wind Eng Ind Aerodyn*. 2011;99(6–7):776-785.
20. Kanamori H, Maechling P, Hauksson E. Continuous monitoring of ground-motion parameters. *Bull Seismol Soc Am*. 1999;89(1):311-316.
21. Shanks JL. Recursion filters for digital processing. *Geophysics*. 1967;32(1):33-51.

How to cite this article: Dan D, Li H. Monitoring, intelligent perception, and early warning of vortex-induced vibration of suspension bridge. *Struct Control Health Monit*. 2022;29(5):e2928. doi:10.1002/stc.2928